# Lesson 2.2 - Quadratic Equations

Department of Mathematics
Visayas State University

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# Objectives

At the end of the lesson, the students will be able to:

- Define quadratic equations.
- Solve quadratic equations using different methods.
- Solve word problems involving quadratic equations.

# Quadratic Equations

**Definition**

A quadratic equation is an equation of the form

$$ax^2 + bx + c = 0$$

where $a, b$, and $c$ are real numbers with $a \neq 0$.

Other common forms of quadratic equations:

- $ax^2 + c = 0$
- $ax^2 + bx = 0$
- $x^2 = c$

Quadratic equations can be solved by factoring, completing the square, and the quadratic formula, and using the following basic property of real numbers.

**Definition (Zero-Product Property)**

$$AB = 0 \text{ if and only if } A = 0 \text{ or } B = 0$$

# Quadratic Equations - Factoring

1. Express the given equation in shorthand form.
2. Factor the left number of the equation.
3. Equate each factor to zero and solve for the resulting equation.
4. Check by substituting the solution of the variable in the given equation.

**Example**

Solve the following quadratic equation by factoring.

1. $x^2 - x = 6$
2. $14x - 11x^2 = 3$
3. $3x^2 - 18x = 0$
4. $a^2 + 5a + 6 = 0$
5. $\sqrt{y-5} + \sqrt{y} = 5$
6. $\dfrac{1}{2x-1} - \dfrac{1}{2x+1} = \dfrac{1}{4}$
7. $x^4 - 9x^2 + 8 = 0$ (reducible)
8. $x - 3\sqrt{x} - 4 = 0$ (reducible)
9. $x^2 - 6x + 9 = 0$
10. $x^2 + 7 = 0$

# Quadratic Equations - Completing the square

This method will make use of the concept of perfect trinomial square and finding the missing term of the perfect trinomial square. The following steps will illustrate the process.

1. The equation should be written in the form $ax^2 + bx = c$ and divide both members by the coefficients of $x^2$ if $a$ is other than 1.
2. Take the coefficient of $x$ and square it. Add this number to both members of the equation.
3. Factor the left side. Then, take the square root of both members and write $\pm$ sign before the right member.
4. Solve the two linear equations.
5. Check the result.

**Example**

Find the roots by completing the square:

1. $a^2 - 12a + 11 = 0$
2. $2x^2 + 5x = 12$
3. $10r^2 = -19r - 6$
4. $3a^2 + 7a = 5$

# Quadratic Equations - Quadratic Formula

**Theorem**

*The roots of the quadratic equation* $ax^2 + bx + c = 0$, *where* $a \neq 0$, *are*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

**Example**

Find the roots using the quadratic formula:

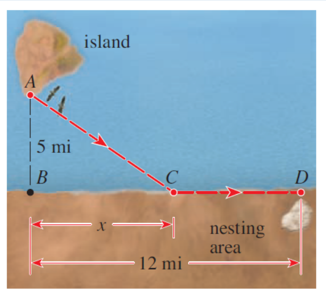
1. $3y^2 - 5y = 2$
2. $2a^2 + a - 15 = 0$
3. $6x^2 + 7x - 20 = 0$
4. $4y^2 + 16y + 15 = 0$

## Example (Verbal Problems)

1. The square of a number $y$ is 90 more than the number. Find the number.
2. The current of a river is 1 km/hr. If a man takes 8 hrs more to row 12 km upstream than to row downstream at the same distance. Find his rate in still waters.
3. Because of an anticipated heavy rainstorm, the water level in a reservoir must be lowered by 1 ft. Opening spillway A lowers the level by this amount in 4 hours, whereas opening the smaller spillway B does the job in 6 hours. How long will it take to lower the water level by 1 ft if both spillways are opened?
4. A jet flew from New York to Los Angeles, a distance of 4200 km. The speed for the return trip was 100 km/h faster than the outbound speed. If the total trip took 13 hours of flying time, what was the jetâs speed from New York to Los Angeles?

## Example (Verbal Problems)

Ornithologists have determined that some species of birds tend to avoid flights over large bodies of water during daylight hours because air generally rises over land and falls over water in the daytime, so flying over water requires more energy. A bird is released from point $A$ on an island, 5 mi from $B$, the nearest point on a straight shoreline. The bird flies to a point $C$ on the shoreline and then flies along the shoreline to its nesting area $D$, as shown in the figure. Suppose the bird has 170 kcal of energy reserves. It uses 10 kcal/mi flying over land and 14 kcal/mi flying over water.



**Example (continuation)**

1. Where should point $C$ be located so that the bird uses exactly 170 kcal of energy during its flight?
2. Does the bird have enough energy reserves to fly directly from $A$ to $D$?